

➤ Document Purpose

This document explains the procedure to upgrade a Red Hat Enterprise Linux system from **RHEL 9.8 to RHEL 10.2** using the Leapp upgrade tool.

➤ Prerequisites

Before starting the upgrade:

- Take a full backup / VM snapshot
- Confirm system has enough disk space
- Verify application compatibility
- Ensure system is registered with Red Hat Subscription Manager
- Take downtime approval

1: Check Current OS Version

Login to the server and verify the current RHEL version.

```
# cat /etc/redhat-release
```

Example Output:

```
Red Hat Enterprise Linux release 9.8
```

2: Update Existing RHEL 9 System

Update all installed packages to the latest RHEL 9 release.

```
# dnf update -y
```

After update completion, reboot the server.

```
# systemctl reboot
```

Verify kernel:

```
# uname -r
```

3: Enable Required Repositories

Enable BaseOS and AppStream repositories required for the upgrade.

```
# subscription-manager repos --enable rhel-9-for-x86_64-baseos-rpms  
# subscription-manager repos --enable rhel-9-for-x86_64-appstream-rpms
```

Verify enabled repositories:

```
# subscription-manager repos --list-enabled
```

4: Install Leapp Upgrade Utility

Install Leapp package.

```
# dnf install leapp-upgrade -y
```

Verify installation:

```
#rpm -qa | grep leapp
```

5: Run Pre-Upgrade Check

Run the Leapp pre-upgrade assessment.

```
# leapp preupgrade
```

Review the generated report.

```
# less /var/log/leapp/leapp-report.txt
```

Check for:

- Inhibitor errors
- Missing packages
- Unsupported drivers
- Configuration issues

Fix all reported issues before continuing.

6: Start RHEL Upgrade

Start the in-place upgrade process.

```
# leapp upgrade
```

If successful, Leapp creates an upgrade boot entry.

7: Reboot Server

Reboot the system to start the upgrade environment.

```
# systemctl reboot
```

During reboot:

- System boots into Leapp upgrade environment
- Packages are upgraded
- System migrates from RHEL 9 to RHEL 10

Wait until upgrade completes.

8: Verify Upgrade

After login, verify OS version.

```
# cat /etc/redhat-release
```

Expected Output:

```
Red Hat Enterprise Linux release 10.2
```

Check running kernel:

```
# uname -r
```

Check failed services:

systemctl --failed

➤ Post Upgrade Validation

- **Check Disk Space**

df -h

- **Check Network**

ip a

ping www.google.com

- **Check Running Services**

systemctl status service-name

- **Check Installed Packages**

rpm -qa

➤ Rollback Plan

If upgrade fails:

1. Restore VM snapshot / backup
2. Boot previous RHEL 9 environment
3. Analyze logs:

#cat /var/log/leapp/

➤ **Important Notes**

- dnf update updates only RHEL 9 packages.
- Major version upgrade (RHEL 9 → RHEL 10) requires Leapp.
- Always test upgrade in a non-production environment first.

➤ **Upgrade Path:**

RHEL 9.8

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| Leapp Upgrade

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RHEL 10.2